



Quantum federated learning: a comprehensive literature review of foundations, challenges, and future directions

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Abstract

Federated learning (FL) is a recent technique that emerged to handle the vast amount of training data needed in machine learning algorithms while fulfilling data owners' privacy challenges in such scenarios. Simultaneously, the field of quantum computing (QC), using quantum properties such as entanglement and superposition to perform computation, has experienced exponential growth, theoretically proving to be more efficient in specific machine learning tasks and creating the discipline known as quantum machine learning (QML). Thus, an emerging body of knowledge has started studying the combination of these two research agendas, giving rise to the field of quantum federated learning (QFL). In this review, we systematically classify the existing literature through a novel taxonomy, identify current trends and challenges, and highlight research gaps and future directions to support the continued development of this emerging field.

Keywords Quantum federated learning · Literature review · Machine learning · Quantum computing

1 Introduction

In today's digital age, the amount of data transferred and generated daily is unprecedented, mainly due to the proliferation of many technological trends such as smartphones, IoT devices, social media platforms, or online services (Oussous et al. 2018).

The increase in the size of the Internet and its associated growth in data usage presents, in general, a positive advantage for artificial intelligence and data-driven machine learning techniques (Khan et al. 2021).

However, two main problems arise. On the one hand, centralized machine learning models may encounter limitations when dealing with huge datasets or with data located at multiple locations. On the other hand, centralized

models introduce the inherent risk of data leakage (McMahan et al. 2017). Even data integration between different departments of the same company faces significant resistance due to industry competition, privacy security, and complicated administrative procedures (Yang et al. 2019). Thus, emphasis on data privacy and security has become necessary to enhance the power of artificial intelligence and data-driven machine learning techniques.

In order to overcome these challenges, McMahan et al. (2017) recently introduced the concept of federated learning (FL). The core concept of this approach is to separate model training from the requirement of direct access to raw training data. This enables distributed computation while significantly reducing privacy and security risks.

Parallely, during the last decades, researchers have strongly investigated how to take advantage of quantum properties such as entanglement or superposition to help improve classical machine learning algorithms. This intersection gave rise to the field known as quantum machine learning (QML), and theoretical speedups for specific tasks have already been published (Liu et al. 2021; Zhang et al. 2024; Shor 1997). At the same time, safer and more secure communications using quantum algorithms have also been theoretically designed (Bennett and Brassard 2014; Ekert 1991). However, current quantum hardware is not yet prone to these implementations (Biamonte et al. 2017; Preskill 2018).

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Due mainly to the theoretical advantage of private and secure communications in the quantum realm but also to the need for distributed quantum computing within the current quantum hardware, the intersection between quantum computing and federated learning, i.e., quantum federated learning (QFL), arises naturally (Chen and Yoo 2021; Li et al. 2021).

As a rapidly emerging field, there is a pressing need for a structured overview to consolidate the diverse and fast-growing body of research. Existing works are scattered across various subfields with inconsistent terminology and approaches, which creates barriers for both new and experienced researchers. This review seeks to fill that gap by providing a clear, exhaustive, and organized foundation to support future advancements in QFL. Our contributions are the following:

- (i) We perform a methodological search, aiming to cover all articles that have focused on QFL so far, and we present this search explicitly to facilitate reproducibility and transparency.
- (ii) We conduct statistical analyses of the reviewed articles to highlight hot topics and the trending growth of the field, among other insights.
- (iii) We propose a novel taxonomy to organize the articles in a visual and intuitive manner.
- (iv) Following this taxonomy, we identify trends, challenges, and gaps in the literature, and we propose several future research directions.
- (v) Lastly, we also identify publicly available software related to QFL.

The structure of this review follows a conceptual framework that moves from foundational knowledge to focused thematic analysis. Sections 2 and 3 provide foundational concepts in FL and quantum computing (QC), respectively. Section 4 outlines our methodology for article selection and presents a statistical overview of the reviewed publications. In Section 5, we expose other literature reviews carried out on QFL to the present time. In Section 6, we present our taxonomy, classifying the literature into three primary categories: efficiency, security, and implementation. Each subsection explores the main contributions, challenges, and future directions within its respective category. In addition, we dedicate a subsection to QFL applications in healthcare, emphasizing FL's particular relevance in this field to address privacy-sensitive data challenges. Finally, Section 7 provides a summary of conclusions drawn from this literature review.

2 Basic concepts in federated learning

This section aims to provide a qualitative and mathematical description of the essence of federated learning. We elaborate

on the pioneering work of McMahan et al. (2017), who introduced the concept.

Generally speaking, a FL structure, represented in Fig. 1, comprises two main components: the central server, shown in the upper part of the figure, and the clients, located at its bottom.

The clients are typically a set of institutions that want to take advantage of a collaborative training model but do not want to share their local data explicitly. For instance, we can think of a set of hospitals that want to benefit from a shared inference model but cannot distribute patients' information due to privacy concerns. This is accomplished by establishing a communication channel between the central server and the clients. In this way, the shared machine learning model is trained locally in each client, and the updated parameters are sent to the server in order to be aggregated, building a global model.

Mathematically, the goal is to minimize an objective function which we assume to be decomposable, and with model parameters w :

$$\min_{w \in \mathbb{R}^d} f(w), \quad \text{where } f(w) := \frac{1}{n} \sum_{i=1}^n f_i(w), \quad (1)$$

that is, we assume a finite-sum objective.

Given K clients, each holding a partition of the data, where P_k denotes the set of indices for data points on client k and $n_k = |P_k|$ represents the number of data points for that client, the objective function in Eq. 1 can be reformulated as follows:

$$f(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w), \quad \text{where } F_k(w) = \frac{1}{n_k} \sum_{i \in P_k} f_i(w) \quad (2)$$

represents the per-datapoint loss function for client k .

The protocol goes as follows. First, the server is instantiated with general model parameters, and a random fraction of clients is selected. The server then transmits the current global model parameters to the selected clients, which in turn use these parameters and their own local datasets to perform local computations.

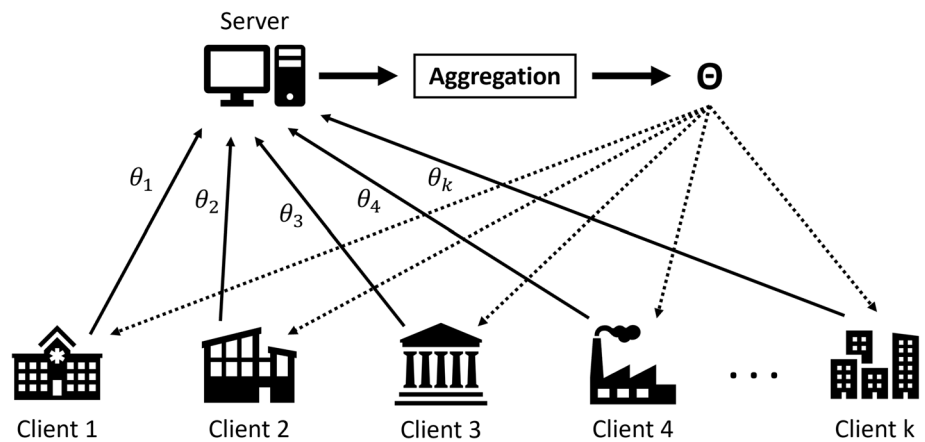
More specifically, with a fixed learning rate η , each selected client k calculates the average gradient on its local dataset, $g_k = \nabla F_k(w_t)$, using the current model w_t .

These gradients are then returned to the central server, which aggregates them. Specifically, the server performs the update

$$w_{t+1} \leftarrow w_t - \eta \sum_{k=1}^K \frac{n_k}{n} g_k, \quad (3)$$

since $\sum_{k=1}^K \frac{n_k}{n} g_k = \nabla f(w_t)$.

Fig. 1 FL typical scheme



The server now sends the updated global parameters to the newly selected clients, and the process starts again, repeating it until a given convergence criterion is met.

On the other hand, an equivalent and more convenient update can be achieved by writing

$$w_{t+1} \leftarrow \sum_{k=1}^K \frac{n_k}{n} w_{t+1}^k, \quad \text{where } w_{t+1}^k \leftarrow w_t - \eta g_k, \quad \forall k. \quad (4)$$

In other words, each client updates the current model by performing a single gradient descent step on its local data, after which the server computes a weighted average of the updated model parameters from all clients.

This is much more convenient if we want to increase the computations because now each client can perform multiple local iterations of the form $w^k \leftarrow w^k - \eta \nabla F_k(w^k)$ before the averaging step. In contrast, if Eq. 3 is used, the number of communications between clients and the server is much larger if we aim to increase computations.

Thus, the protocol is defined by three key parameters: the number of clients participating in training each round, the number of training epochs each client completes over its local dataset per round, and the minibatch size each client uses for its updates.

2.1 Federated learning classification

From a qualitative point of view, Yang et al. (2019) proposed a classification of FL depending on the characteristics of the data distribution. In short, one can talk about horizontal federated learning (HFL), where clients' datasets share the same feature space but have different samples; vertical federated learning (VFL), where clients' datasets share the same sample space but have different feature space; and federated transfer learning (FTL), where clients' datasets differ not only in samples but also in feature space (Yang et al. 2019). To illustrate these differences, we could

imagine a hypothetical situation with a set of hospitals, as shown in Fig. 2. If each hospital has similar (ideally the same) types of data about different patients (e.g., demographic information, medical history), we talk about HFL, represented in Fig. 2a. If a hospital has only clinical records, and another has only health monitoring data (e.g., physical activity, vital signs, etc.) regarding similar (ideally the same) patients, we talk about VFL, represented in Fig. 2b. Finally, if two hospitals have different types of data regarding different patients, they can still try to train a global model, and we talk about FTL, represented in Fig. 2c.

Despite being first proposed in 2017, FL has gained much attention both from academia and industry. This field already finds applications across multiple sectors, including health-care, education, data science, and finance, among many others. Moreover, it is often implemented with various technologies such as artificial intelligence, Internet of Things devices, natural language processing, blockchain technology, or autonomous vehicles, to mention just a few (Banabilah et al. 2022; Li et al. 2020; Shaheen et al. 2022).

Interested readers can find more information about FL in Guendouzi et al. (2023), which offers an exhaustive and systematic review of the field, emphasizing its categories, challenges, aggregation techniques, and associated development tools; in Shaheen et al. (2022), a systematic review which focuses mainly in its application to different industries and domains; or in Kairouz et al. (2021), which focuses on challenges and open problems, to cite a few examples.

3 Quantum computing

Steane (1998) described the concept of quantum computing as the field that results from the intersection between classical information theory, computer science, and quantum physics.

Yet, quantum computing's roots can be traced further back in the past to the pioneering paper by Feynman (1982) where

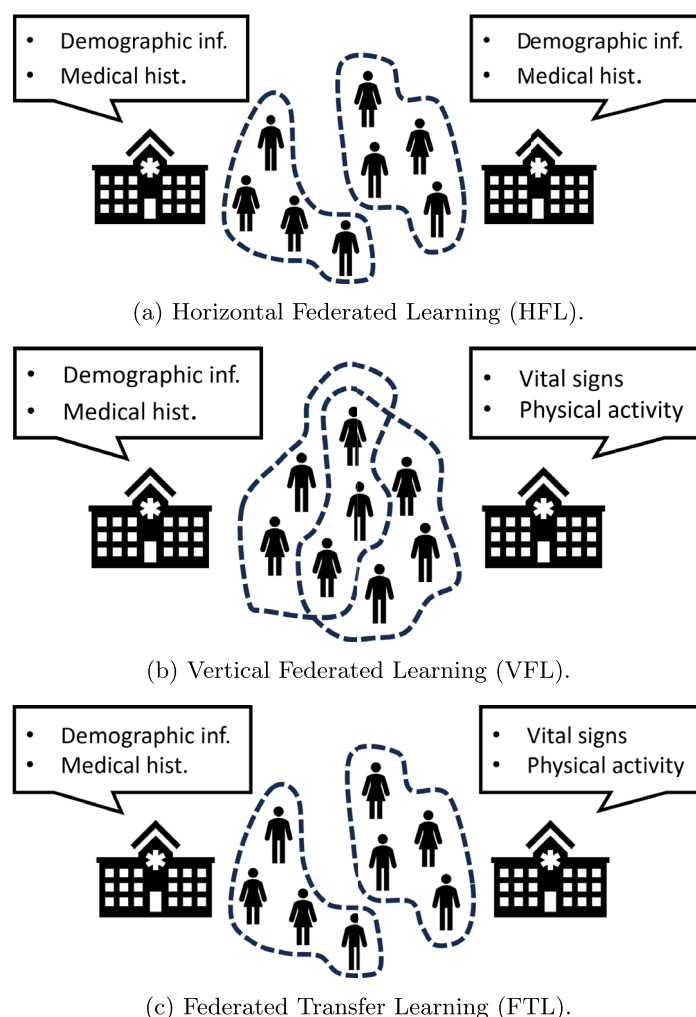


Fig. 2 Classification scheme of FL

the author suggested that certain quantum phenomena, such as entanglement or superposition, could efficiently simulate physical processes, a task that classical computers struggled with. From that moment onward, speculations arose regarding the use of such quantum properties to speed up computation in general, leading to the development of faster algorithms, novel cryptography mechanisms, and improved communication protocols (Rieffel and Polak 2011).

The work from Shor (1997), proving that a quantum computer could efficiently factor large numbers, or the one from Grover (1996), showing that quantum computers could search an unsorted database quadratically faster than classical algorithms, marked a significant theoretical milestone in the field of quantum computing. These advances and the extended growth of data-driven machine learning techniques gave rise to the field known as quantum machine learning (QML), which was thought to be a probable candidate to potentially revolutionize future ways of intelligent data processing (Schuld et al. 2015).

3.1 Insights on quantum machine learning

Schuld and Petruccione (2018) classified the intersection between quantum computing and machine learning into four categories. Depending on the type of device used (either a quantum or a regular computer) and the type of data processed (either classical or derived from a quantum system), the authors distinguish between *quantum for classical*, i.e., processing classical data on a quantum computer; *classical for quantum*, i.e., processing quantum data on a classical computer; *quantum for quantum*, i.e., processing quantum data on a quantum computer; or *classical for classical*, i.e., processing classical data on a classical computer, though in this case, with the use of *quantum-inspired* algorithms, such as tensor networks. However, as the authors of this work point out, the main focus of QML is on the *quantum for classical* setting with the idea of designing quantum algorithms to process classical datasets, such as text, images, or time series of classical systems, using a quantum computer.

Thus, researchers increasingly focused on pursuing the so-called quantum advantage, also known as *quantum supremacy*, by proving the capability of a programmable quantum computer to solve a problem beyond the reach of any classical computer within a feasible amount of time (Preskill 2012). In 2019, Google claimed to have achieved the long-awaited quantum supremacy by solving a particular problem in a few hundred seconds with their 53-qubit processor and stating that the equivalent task for a state-of-the-art classical supercomputer would take approximately 10,000 years (Arute et al. 2019). In response, IBM expressed that no such claim could be stated and listed many techniques a supercomputer could do to solve that task in no more than 2.5 days (Zialcita 2019).

Since those days, the same situation has been encountered repeatedly: with every claim of quantum supremacy, scientists have designed novel and more efficient classical algorithms that seem to challenge this primacy (McCormick 2022).

All in all, quantum supremacy has not been proved yet, and a debate on whether or not research lines should focus mainly on surpassing classical computers has recently started (Schuld and Killoran 2022).

One of the primary factors contributing to this is the limitation of the current quantum hardware. As Preskill (2018) expressed, we are now navigating the Noise Intermediate-Scale Quantum (NISQ) era, a transitional period in which the size of quantum computers will be no more than a few hundred qubits and will present control imperfections.

3.2 Quantum computing approaches

Despite today's imperfect quantum computers, various approaches and technologies are being developed to pave the way for future fault-tolerant quantum machines.

The most popular approach is the use of quantum circuits built with the help of quantum gates (Schuld and Petruccione 2018). Analogously to a classical electric circuit, quantum gates are parametrized electromagnetic interactions that can be used to manipulate a single qubit or multiple qubits simultaneously. Within this model, different hardware techniques are being researched, such as using polarized photons, trapped ions, or superconducting materials, to name a few (Ladd et al. 2010). Moreover, with the limitations of the actual hardware, different methods have emerged to leverage the advantages of quantum circuits. An example is the application of optimal control theory, which focuses on optimizing a continuous laser pulse, summarizing an entire gate sequence (Glaser et al. 2015). Another example is variational quantum algorithms (VQA), which use a classical optimizer to train a parametrized quantum circuit (Cerezo et al. 2021). Under this vast term, one can find highly used algorithms such as a quantum neural network (QNN), a quantum

convolutional neural network (QCNN), or a variational quantum circuit (VQC) in general. We will not elaborate any further on VQA, but most knowledge can be found in Kwak et al. (2022) or in Ren et al. (2025).

An entirely different approach to quantum computing is the so-called *quantum annealing*, which at the same time can be understood as a heuristic to *adiabatic quantum computing* (Schuld and Petruccione 2018). This approach suggests encoding the solution of the optimization problem into the ground state of a Hamiltonian that characterizes the dynamics of the n -qubit system. Starting from the ground state of another more easy-to-prepare Hamiltonian and slowly adjusting the system to follow the desired dynamics, the system is expected to remain in the lowest energy state, solving the initial optimization problem (Farhi et al. 2000). In 2011, the Canadian company *D-Wave* was the first to commercialize a quantum annealer (Johnson et al. 2011).

3.3 Quantum software libraries

Despite the early stage of development for quantum hardware, there has been significant work on quantum programming languages (Garhwal et al. 2021). In fact, quantum programming languages have been around even before the development of real quantum devices, simulating quantum algorithms on classical computers since the early days (Cervera-Lierta 2018). As companies start building and commercializing their own quantum hardware, they are also starting to develop their own quantum programming language in order to use their infrastructure. This has resulted in a variety of languages that tackle many different levels of programming, ranging from assembly languages, such as *OpenQASM*, an instruction language for *IBM*'s quantum machines, to higher-level languages, such as *Q#*, a quantum development toolkit from *Microsoft* where quantum algorithms are already programmed at a low level (Heim et al. 2020).

Next, we present some quantum programming languages (for a full list, check the [Open-Source Quantum Software Projects](#) catalog):

- **Qiskit**: The Quantum Information Software Kit, or Qiskit, is the open-source quantum computing software development framework from *IBM*. It allows one to work, mainly in Python, on various programming levels, from working on pulses, gates, and circuits to simply calling and running quantum algorithms. It offers both a cloud quantum computing service and a quantum simulator.
- **Cirq**: Cirq is an open-source Python library from Google mainly used for creating and editing quantum circuits. It offers the possibility of running code in cloud quantum computers or locally in a simulator.

- **Strawberry Fields** and **Blackbird**: These two Python libraries from the Canadian company Xanadu Quantum Technologies are often used to design, simulate, and optimize quantum optical circuits in Xanadu's photonic quantum computers.
- **PennyLane**: Apart from the above two, Xanadu also developed PennyLane, a cross-platform Python library for quantum machine learning and differentiable programming of quantum computers.
- **Ocean**: Ocean is a suite of open-source Python tools to run on D-Wave quantum computers, a Canadian company focused on developing adiabatic quantum computers. With the use of other D-Wave native Python libraries such as *dwave-system* or *dimod*, users can solve optimization problems by mapping them into Ising Hamiltonians and running them on quantum annealers.
- **Forest**: Forest is a developer environment offered by the American company Rigetti Computing. It allows one to write and run quantum programs by means of a lower-level language called *quil* or through a Python library called *pyquil*. In addition, they have also developed another library called *Grover*, which includes several quantum algorithms for optimization techniques.
- **Q#**: Microsoft does not yet have a publicly available quantum computer, but they do offer a set of tools, such as the Quantum Development Kit, which includes their novel quantum programming language Q# and a set of libraries with a Visual Studio environment integration to run quantum algorithms on a simulator.

4 Methodology

In this section, we outline the methodology employed to search for and filter relevant works related to QFL. Additionally, we carry out a comprehensive statistical analysis of the selected articles, offering insights into the trends and focus areas within the QFL literature.

4.1 Search procedure

The article search was performed in November 2024, taking into account all papers published until that date. We used two different approaches.

On the one hand, we used the Boolean clause with the search strings “quantum federated” OR “federated quantum” to look for articles published in *Springer*, *IEEE Xplore*, *ACM Digital Library*, *Science Direct*, and *Google Scholar*.

On the other hand, we performed on the same platforms title-specific searches, that is, the search was performed only within articles' titles, using the terms “federated” AND “quantum” and “federated” AND “quantumvolutional.”

By doing so, we obtained 28 articles on *Springer*, 65 on *IEEE Xplore*, 13 on *ACM Digital Library*, 12 on *Science Direct*, and 520 on *Google Scholar*, summing up to a total of 638 articles.

4.2 Selection procedure

To guarantee the relevance of the articles chosen for this study and enhance the rigor and accuracy of the selection procedure, two independent reviewers carried out a detailed review process. Initially, the title, keywords, and abstract of each article were evaluated to assess its suitability for inclusion. Moreover, in cases where further clarification was required, a full-text review of the article was performed to determine its relevance. Articles not written in English were directly excluded. Following this, duplicate papers were identified and excluded from further analysis. After this procedure, shown in Fig. 3, we ultimately identified a total of 103 articles regarding QFL.

4.3 Statistical analysis

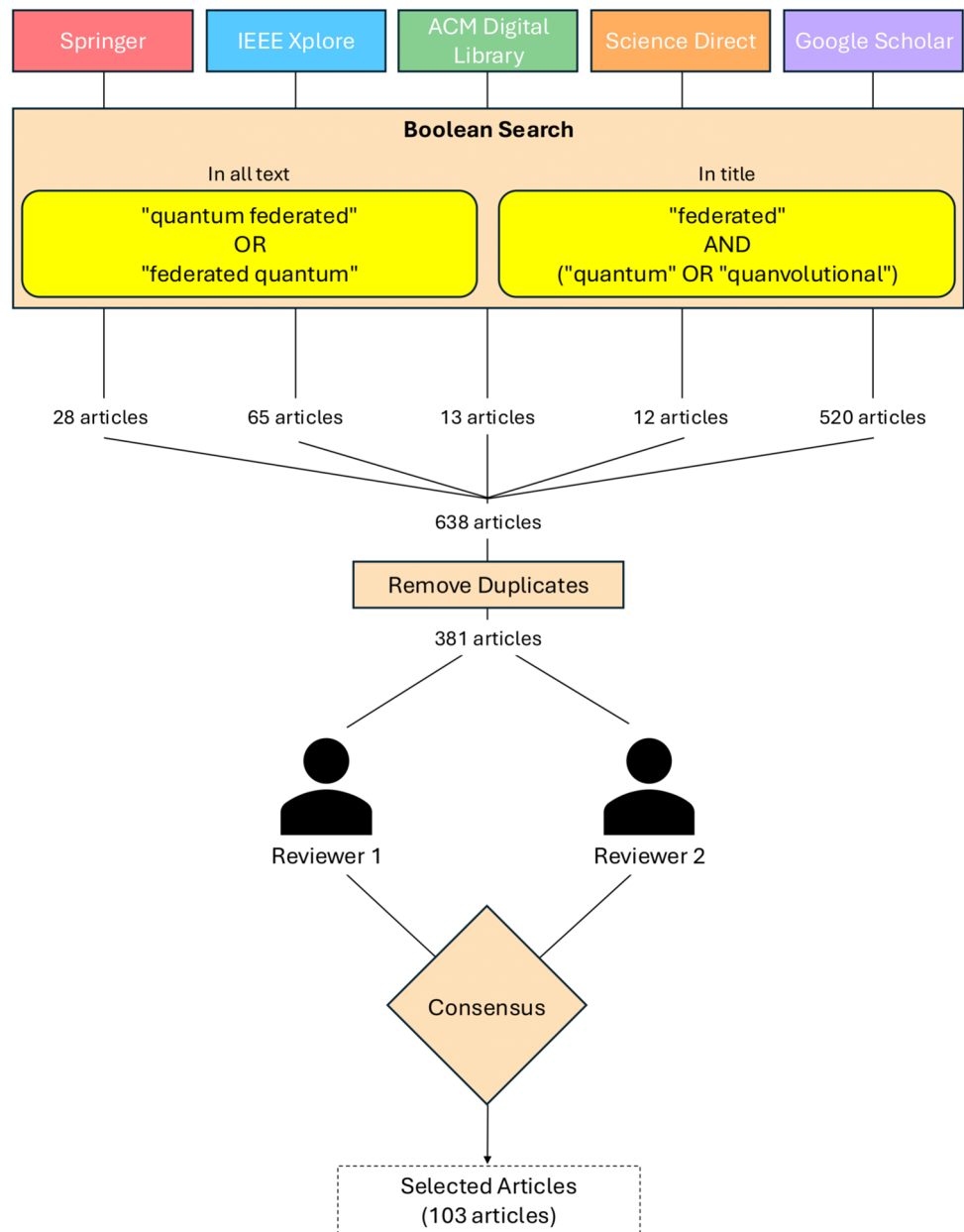
In this section, we statistically analyze the final papers resulting from the selection procedure, which will represent the scope of our study. The goal of this analysis is to allow us to discern temporal trends in the evolution of research, evaluate the impact and influence of these papers, and explore potential patterns or disparities in research contributions based on the geographical origin of the primary authors, among others.

Figure 4 illustrates the increasing trend in the number of articles published each year in the field of QFL. The data reveals a remarkable upward trajectory, underscoring the burgeoning interest in this innovative intersection of quantum computing and FL. In 2020, the field saw its inaugural publication, with just one article (Narottama and Shiny 2020). This initial interest laid the groundwork for subsequent research, as the potential applications and implications of QFL began to capture the attention of scholars and practitioners alike. The following year, in 2021, the number of publications surged to 8, indicating a growing recognition of QFL's relevance in the broader landscape of machine learning and quantum technologies.

As we move further into the timeline, 2022 marked another significant leap, with 11 papers published. This steady increase reflects not only a rising interest but also the establishment of foundational frameworks and theoretical discussions around QFL. By 2023, the growth became even more pronounced, with a staggering 28 articles published—a significant jump that demonstrates the field's accelerating momentum.

The trend is evidenced by the actual data for 2024, which shows a remarkable increase to 55 published articles. This sharp rise indicates that the QFL research community is not

Fig. 3 Article selection flowchart. We used Boolean queries across five academic platforms, first targeting the entire text with the query “quantum federated” OR “federated quantum” and then with two title-specific searches: one for “federated” AND “quantum” and another for “federated” AND “quantumvolutional.” After deduplication, two independent reviewers assessed each article for selection or rejection, resulting in a final selection of 103 articles for review



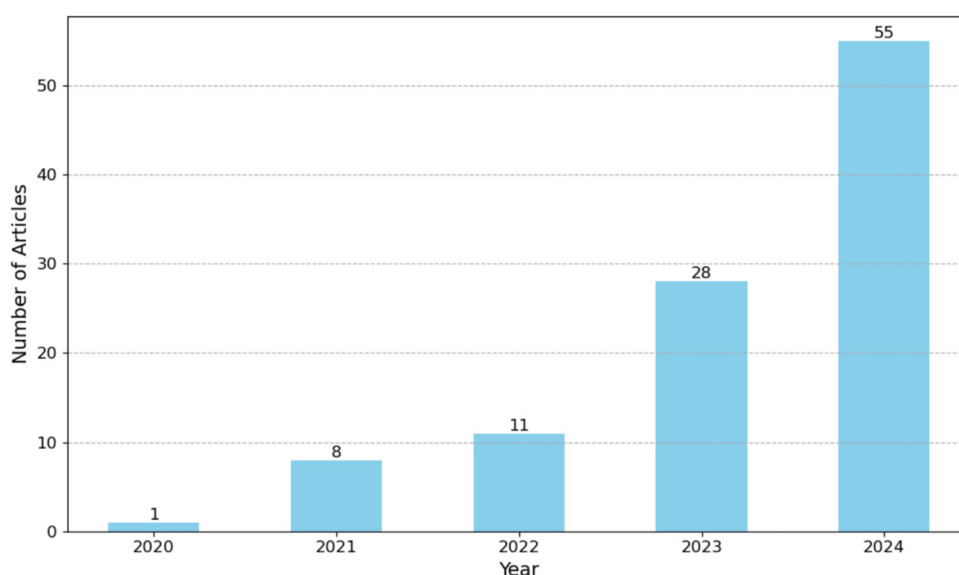
only expanding but also becoming increasingly active and collaborative. The nearly doubled output compared to the previous year in 2024 represents an astonishing 53% of the total articles reviewed, further solidifying QFL’s position as a key area of focus in both academia and industry.

As the fields of QML and FL evolve and intertwine, the relevance of QFL will likely continue to grow. Researchers are beginning to explore novel methodologies, applications, and theoretical underpinnings that harness the advantages of quantum mechanics while addressing the challenges associated with FL. All in all, the trajectory of published articles in the field of QFL highlights a significant trend, showing that more people are realizing its importance and how it could change our approach to FL in the quantum era.

Figure 5 presents a pie chart that illustrates the countries affiliated with the institutions of the articles’ first authors, highlighting the percentage of contributions from each nation on QFL. The data reveals that the USA is the leading nation in this emerging field, with a total of 32 publications. This significant output underscores the USA’s prominent role in advancing QFL research and development.

Following the USA, China emerges as the second most prolific contributor, with 18 articles. Notably, when aggregating the contributions from both China and the Republic of Korea, the total number of publications reaches 37. This combined output highlights a burgeoning interest and commitment to QFL research within these Asian nations, indicating a collaborative and competitive landscape in this

Fig. 4 Total number of papers published each year on QFL based on articles identified through the selection process described in Fig. 3, up to November 2024



domain. Furthermore, both Singapore and India have made substantial contributions, with 8 articles attributed to each country. This trend indicates a significant focus on QFL research among Asian countries, which are increasingly establishing themselves as key contributors in the global academic landscape of quantum technologies.

In stark contrast, the representation of European countries in the QFL literature appears relatively limited. Collectively, France, Germany, Greece, Italy, and the UK contribute only 8 articles. This low representation raises questions about the engagement of European researchers with QFL and suggests potential areas for growth and collaboration.

Additionally, Australia has produced a noteworthy total of 6 articles, with four of these contributions originating from the School of Information Technology at Deakin University, Victoria (Pokhrel et al. 2024; Gurung et al. 2023b, a; Gurung and Pokhrel 2024).

Overall, the data underscores a significant research focus on QFL from the USA and various Asian nations, while contributions from European countries and other regions—including Morocco, Nigeria, Taiwan, Turkey, and Bangladesh—remain minimal. This disparity suggests a need for increased international collaboration and engagement to enhance the global discourse on QFL and its potential applications.

The pie chart in Fig. 6 illustrates the distribution of publication types among the reviewed articles, categorized into journals, conferences, and open-access platforms such as *arXiv*. Nearly half (47.9%) of the articles were published in peer-reviewed journals, making this the largest category. This suggests a strong emphasis on high-quality, validated research, as journals are known for maintaining rigorous publication standards.

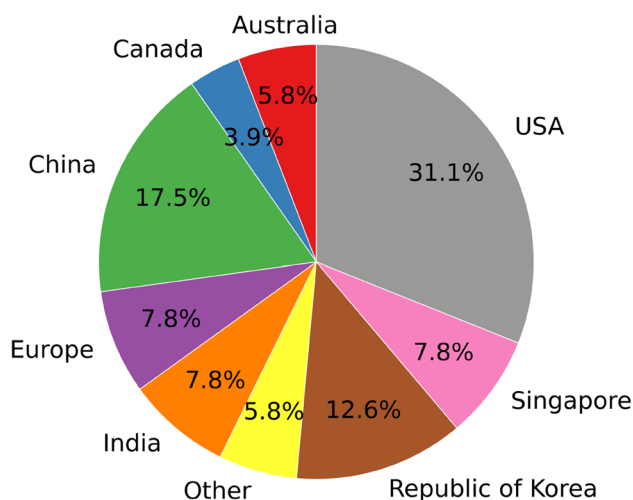


Fig. 5 Distribution of first authors' institution country

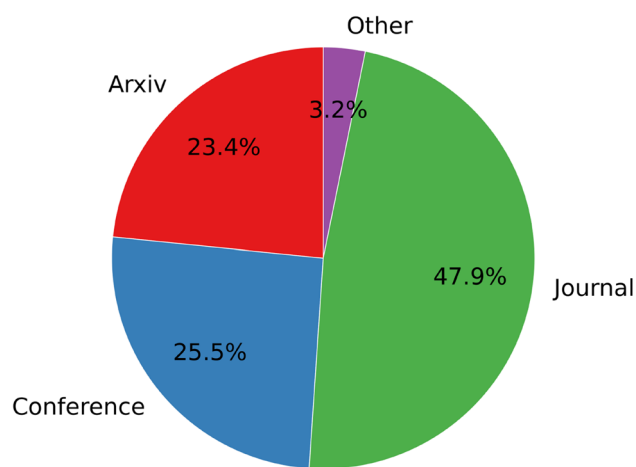


Fig. 6 Distribution of articles by publication type

Following journals, 25.5% of the articles were presented at conferences, highlighting the role of conferences as a key venue for the timely dissemination of emerging research. This substantial proportion reflects how conferences enable the rapid exchange of ideas, allowing researchers to share preliminary findings with the academic community.

The remaining 26.6% of articles were published on arXiv or similar open-access platforms. This significant presence reflects the recent emergence of QFL as a field, where researchers often seek to publish quickly to share findings with the global community, even as many pursue or await peer-reviewed publication. This approach supports timely access and collaboration, which are essential in rapidly developing research areas.

Overall, this distribution shows a balanced approach in the research field, with the community leveraging both traditional peer-reviewed venues and open-access platforms to support rigorous review and widespread, timely dissemination of findings.

Among the most popular venues of publication, we find the following journals: Quantum Machine Intelligence, Science China Physics, Mechanics & Astronomy, The Journal of Supercomputing, IEEE Internet of Things Journal, and Quantum Science and Technology, and the following conferences: IEEE International Conference on Quantum Computing and Engineering (QCE), International Conference on Computing, Networking and Communications (ICNC), International Conference on Quantum Communications, Networking, and Computing (QCNC), International Conference on Information and Communication Technology Convergence, IEEE International Conference on Cyber Security and Cloud Computing (CSCloud). We observe that the distribution between quantum and classical journals or conferences is fairly balanced. This might suggest that the field of QFL is actively bridging the gap between classical machine learning and

quantum computing, with contributions coming from both established classical venues and emerging quantum-focused platforms.

Lastly, Figs. 7 and 8 aim to highlight the key themes and areas of focus in the field. On the one hand, Fig. 7 displays a word cloud representing the 50 most repeated keywords across the reviewed articles. The size of each word in the cloud is proportional to its frequency of occurrence, aiming to provide a visual summary of the most common themes and topics within the articles. On the other hand, Fig. 8 shows the distribution of articles according to their application or key topic. The most popular category is the *Classification Task*, which encompasses works that do not concentrate on specific real-world applications but instead focus on improving the performance of classifier models, frequently utilizing well-established datasets such as MNIST. The second-largest category, *IoT Ecosystems*, includes articles that apply and enhance QFL in the context of interconnecting various devices, as seen in applications such as smart cities, satellite-ground communications, or autonomous vehicles applications. Finally, the third-largest category, *Cybersecurity*, comprises articles that focus on advancing QFL techniques for privacy preservation and attack mitigation.

These two figures played a key role in shaping the taxonomy presented in Section 6, helping us identify and define the primary areas of focus within the field.

5 Related work

As seen previously, QFL is a relatively new yet rapidly expanding field. Despite its early stage of development, QFL has attracted growing interest due to its potential to revolutionize decentralized machine learning by leveraging quantum computational power. As research in this area

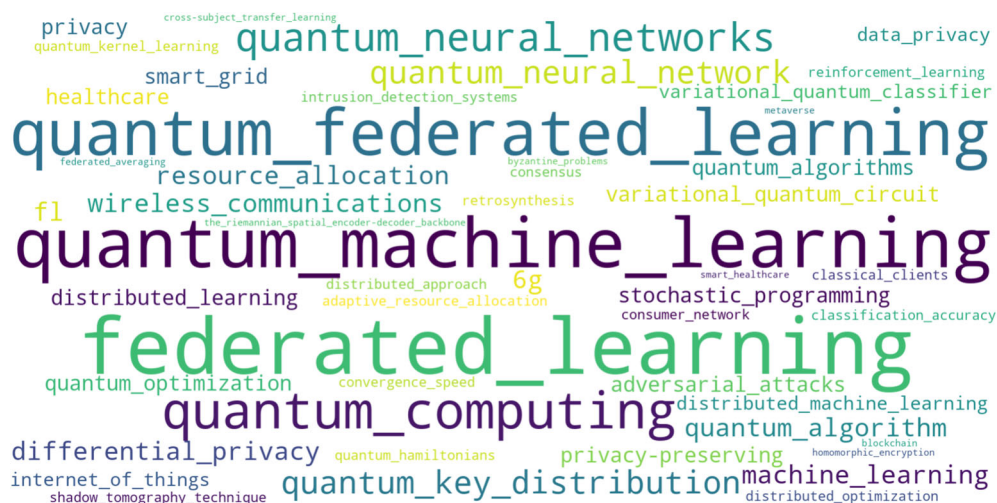


Fig. 7 Word cloud depicting the 50 most commonly repeated keywords found in the articles reviewed, with word size reflecting their frequency

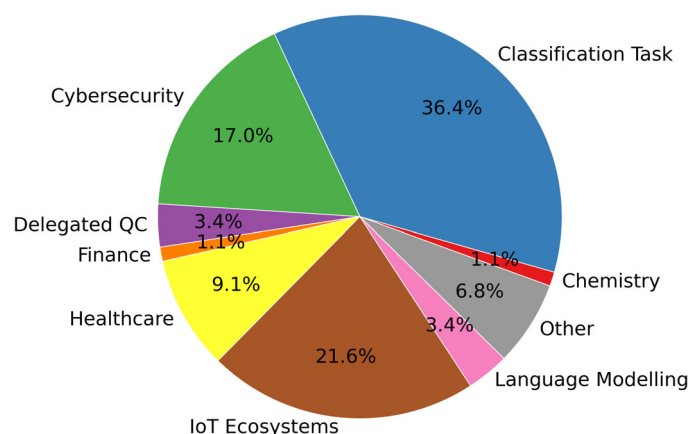


Fig. 8 Distribution of the key focus areas or applications addressed by the reviewed articles. The three main categories encompass *Classification Task*, which focuses on model performance improvement; *IoT*

Ecosystems, which deals with interconnecting devices in applications like smart cities and autonomous vehicles; and *Cybersecurity*, which addresses privacy protection and attack mitigation

accelerates, a few literature reviews have been conducted to assess the field's current progress, challenges, and future directions.

A common trend in articles reviewing QFL is their reference to it in the context of classical FL reviews, especially in the sections focused on future directions. In this way, as researchers explore ways to overcome the limitations of classical FL, such as privacy concerns, scalability, and communication overhead, qfl is frequently mentioned as a promising solution due to the potential quantum computing holds for addressing these issues (Saha et al. 2024; Ren et al. 2024; Ha et al. 2023; Karydas and Leligou 2024; Jagarlamudi et al. 2024).

These reviews typically touch upon the foundational ideas of QFL, highlighting its ability to leverage quantum speed-ups and provide enhanced security. While these mentions are often brief and speculative, they signal the growing recognition of QFL as an important area of research that could play a transformative role in classical FL.

Conversely, there are several reviews on QML that highlight the potential of QFL as a possible approach to enhance privacy and resource distribution. By leveraging FL's ability to train models collaboratively without sharing sensitive data, these reviews emphasize how QFL can be of use when addressing Internet of Things (IoT) environments (Kang and Nam 2024; Zaman et al. 2023; Javeed et al. 2024; Rane et al. 2024) or healthcare applications (Zhou et al. 2024).

On the other hand, there are also a few articles that specifically focus on reviewing the field of QFL.

Larasati et al. (2022) give a short, concise state-of-the-art review for a few papers written mostly in the early years of the field, namely Xia and Li (2021), Narottama and Shiny (2020), Chen and Yoo (2021), Chehimi and Saad (2022),

Li et al. (2021), and Yamany et al. (2023). The article primarily focuses on exploring strategies and potential conditions for deploying QFL, highlighting several challenges, such as the limitations of current quantum hardware or the need to address realistic scenarios.

Similarly, Gurung et al. (2023b) provide a more detailed analysis of the previously mentioned articles and includes additional works such as Huang et al. (2022), Xia et al. (2021), Zhao (2022), Zhang et al. (2023), Gurung et al. (2023c), and Yun et al. (2022). Furthermore, they present two proofs-of-concept: first, they classify the IRIS dataset using a variational quantum classifier within a federated environment, utilizing the PennyLane library. Second, they leverage both PennyLane and the JAX framework to classify three distinct classes from the MNIST dataset, employing a variational quantum circuit and slightly modifying the code from Gurung et al. (2023c).

Kwak et al. (2022) offer another review of the concept of QFL, focusing on the limitations and challenges of quantum distributed deep learning (QDDL), with QFL as the central focus. Their discussion primarily draws on and summarizes the work of Chen and Yoo (2021) and Li et al. (2021), which, to the best of our knowledge, are among the first contributions to the field of QFL. In addition, they provide an overview of potential security measures, such as quantum key distribution (QKD) and quantum secure direct communications (QSDC), and briefly touch on future challenges, including scalability and communication issues.

Analogously, Bu (2021) provides a brief, one-page review focused only on the work by Li et al. (2021). In this review, the author advocates for extending the QFL framework discussed to explore other quantum algorithms, such as quantum support vector machines and generative models.

On a related note, Brei (2022) presents a bachelor's thesis that outlines the fundamentals of machine learning, quantum computing, and QFL, supplemented with experimental work involving a binary classifier.

In contrast, one of the most recent articles reviewing QFL is the one by Chehimi et al. (2024a). However, this work is highly focused on addressing the challenges and potential future directions for deploying QFL over both classical and quantum networks. The authors discuss key issues in network infrastructure and propose strategies to enhance the performance and security of QFL in these environments.

The remaining two most recent reviews regarding QFL are provided by Ren et al. (2025) and Qiao et al. (2025).

Ren et al. (2025) present an extensive review of QFL, with their primary contribution being the introduction of a taxonomy that classifies QFL into two key categories. The first focuses on utilizing the advantages of QC to improve the efficiency of QFL compared to classical FL, while the second emphasizes secure methods to ensure data privacy, model security, and robustness against adversarial attacks. In the first category, they explore concepts such as quantum data encoding (QDE), quantum feature selection (QFS), and various global and local models, including aggregation and client selection strategies. The second category delves into security measures, including QKD, QSDC, quantum secure multi-party communication (QSMPC), quantum oblivious transfer (QOT), homomorphic encryption (HE), and differential privacy (DP), referencing relevant works when possible.

On the other hand, Qiao et al. (2025) offer a comprehensive review of classical FL and key concepts in QML, leading to a smooth transition into QFL. Although the work is relevant and engaging, the section dedicated to QFL is relatively brief. We believe this might be due to the fact that, like the review from Ren et al. (2025), the authors of both works conducted their article search in the first half of 2023.

As shown in Fig. 4, QFL has experienced rapid growth in popularity. In fact, by 2024, more articles had been published

on this topic than in all previous years combined. This highlights the importance of providing an up-to-date, comprehensive review that encompasses the latest research in the field.

Moreover, none of the reviews covered in this section exhibits an explicit, exhaustive search on the subject nor a statistical analysis to evaluate the impact, temporal evolution, or geographical disparities of the reviewed articles regarding QFL.

Table 1 compares our review with all other existing reviews regarding QFL, all of which have been covered in this section. To the best of our knowledge, we are the only authors who have explicitly conducted a systematic search to filter the selected articles and the only ones who have performed statistical analyses of these works. Additionally, only Ren et al. (2025) and our work present a taxonomy for QFL.

6 Results and discussion

Based on our analysis of the existing literature, we propose a novel taxonomy for QFL that classifies the works into three key categories: efficiency, implementation, and security. This framework aims to provide a structured approach to understanding the current landscape of the field, highlighting the critical areas where advancements are being made and identifying key challenges and future directions within each category.

In short, the efficiency category includes papers that focus on optimizing performance. These works typically compare different algorithms and datasets, evaluating a range of metrics such as testing accuracy, computational efficiency, and convergence rates.

The security category, on the other hand, is concerned with analyzing and mitigating various security threats. This includes addressing vulnerabilities to adversarial attacks,

Table 1 Comparison between our and other reviews regarding QFL. We compare whether each review has conducted a methodological search, performed a statistical analysis of the assessed studies, proposed a QFL taxonomy, and discussed applications, challenges, and future directions related to QFL

Review reference	Meth search	Stat analysis	QFL taxonomy	QFL app	QFL challenges	Future directions
Larasati et al. (2022)	—	—	—	—	✓	—
Gurung et al. (2023b)	—	—	—	✓	✓	✓
Kwak et al. (2022)	—	—	—	✓	✓	—
Bu (2022)	—	—	—	—	—	—
Chehimi et al. (2024a)	—	—	—	—	✓	✓
Ren et al. (2025)	—	—	✓	✓	✓	✓
Qiao et al. (2025)	—	—	—	✓	✓	—
Ours	✓	✓	✓	✓	✓	✓

data poisoning, and privacy breaches, as well as exploring quantum and classical encryption techniques, with the aim of ensuring robust and secure model training within a QFL framework.

Finally, the implementation category focuses on the practical aspects of deploying QFL models, aiming to bridge the gap between theoretical research and real-world applications and contributing to making QFL a practical technology in the near future.

Nevertheless, not all papers necessarily fall into just one of these categories. That is, some articles can be found simultaneously in more than one class. For instance, we will show that some publications, such as Narottama et al. (2021), focus on communication optimization, blending elements of both the efficiency and the implementation categories. Other works merge quantum algorithms from an efficiency point of view with classical security methods, such as DP or HE, to enhance robustness against various attacks. These works, like those in Li et al. (2021) and Chu et al. (2023), integrate efficiency and security to create more resilient QFL models.

Thus, we propose Fig. 9 to represent the proposed taxonomy. This diagram allows us to locate papers not only in one of the three categories discussed before but also in mixed classes.

We believe this classification is natural, exhaustive, and more intuitive and pragmatic than the one proposed in Ren et al. (2025). Apart from these two, no other classification has been proposed to the best of our knowledge.

In the following subsections, we will review the selected articles on QFL, organized according to the taxonomy we proposed earlier. Each subsection will focus on the most significant works within that category, providing a

comprehensive discussion of key contributions to the field. In addition, we will highlight emerging trends, identify ongoing challenges, and propose potential future directions for research in each area. Moreover, we have also included a dedicated subsection on articles that apply QFL within healthcare, as FL is particularly well-suited for this domain due to the strict privacy policies and data protection requirements inherent in healthcare.

By structuring the review in this way, we aim to offer a clear and organized overview of the current state of QFL, shedding light on the opportunities and obstacles that researchers are likely to encounter as the field continues to evolve.

6.1 Efficiency

6.1.1 Baseline and literature review

Out of the 103 articles reviewed, 53 focus on efficiency, that is, in achieving similar or better performance compared to other methods, for instance, by comparing their frameworks to the traditional centralized approach.

One of the main goals explored by the works reviewed in this section is to investigate how quantum speedup might benefit QFL. This includes reducing the computational burden on clients, accelerating convergence during training, and enabling more compact model representations. Although these advantages are still largely theoretical or supported primarily through simulations, they serve as a key motivation behind ongoing research in this area.

The article by Chen and Yoo (2021) serves as the baseline for this strategy and, to the best of our knowledge, it is also one of the first articles to introduce the concept of QFL, together with the work by Li et al. (2021).

In this paper, the authors use a VQC instead of a classical neural network and plug it inside the well-known classical FL structure. In this sense, the only difference with FL is that the clients, i.e., each node, now have a quantum computer that optimizes rotation parameters instead of weights and biases of a neural network. As in the classical scenario, these parameters are sent to the server, which computes and sends a global model, aggregating the parameters classically by computing a weighted average.

The authors compare the performance of a non-FL setting with that of a FL setting in the classification of two distinct datasets: one consisting of cat and dog images and the other consisting of car and plane images. They conduct a series of experiments by varying the number of local epochs and demonstrate that they can achieve nearly identical accuracy in both settings while significantly reducing computational costs within the FL framework.

Most of the works aimed at improving performance on various metrics build upon the foundational work of Chen and

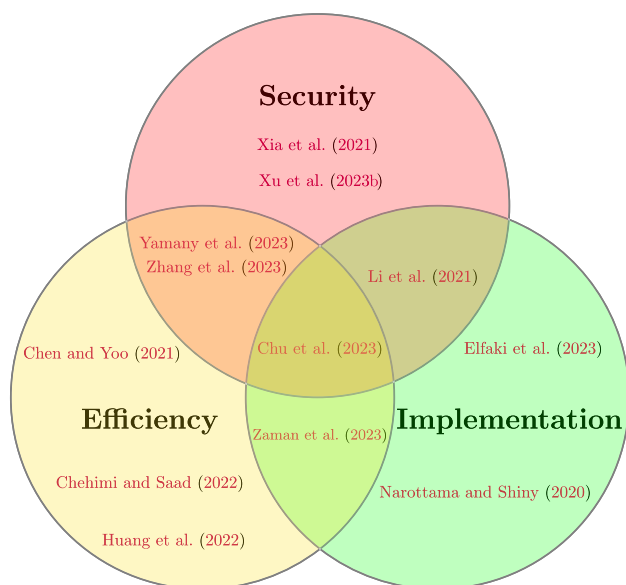


Fig. 9 Proposed QFL taxonomy with some articles for each category

Yoo (2021), reflecting its significant influence in the field. Notably, 48 of these papers, i.e., more than 90% of the works in the efficiency category, focus on performing a classification task.

For instance, the works Sünkel et al. (2024); Lee and Park (2024); Singh et al. (2023); Gurung and Pokhrel (2024); Wang et al. (2024) examine the performance of QFL in classifying the MNIST dataset by varying several model parameters, such as the number of clients, local epochs, or communication rounds, and using VQCs as building blocks for their models, following the work from Chen and Yoo (2021).

Similar studies can be found in Sünkel et al. (2024); Wang et al. (2024); Wu et al. (2024); Qu et al. (2023); Bhatia and Neira (2024) where the authors tackle different binary datasets under very similar frameworks. Qi et al. (2023) also focus on a binary classification but proposes to upload clients' local gradients to the server rather than the actual parameters. Moreover, using natural gradient descent shows how this framework achieves faster convergence rates, lowers communication costs, and maintains baseline performance.

On the other hand, the work from Yun et al. (2022) performs a classification task on 4 classes of the MNIST dataset by optimizing not only the angle parameters of their VQC but also the basis on which they perform the measurements, parametrized with the so-called *pole parameters*. The authors show how this framework performs better than vanilla QFL, especially under poor channel conditions.

Works focused on anomaly detection in various sectors, such as malware detection (Houda et al. 2024), financial fraud (Innan et al. 2024), and healthcare applications (discussed in Section 6.4), utilize nearly identical frameworks.

Remarkably, it is important to note that there are also a few works that differ significantly in their approach and methodology. For instance, Chen (2023) addresses three well-known reinforcement learning problems by using VQC as local agents and training them asynchronously. The author compares this framework to classical reinforcement learning models with similar size and architecture and achieves superior or comparable performance.

Chehimi et al. (2024b) introduce the federated quantum version of the long short-term memory (LSTM) algorithm and is able to successfully approximate three different types of functions. Moreover, numerical simulations show that their QFL framework achieves faster convergence rates compared to classical LSTM models in a traditional FL setup. Similarly, Li et al. (2023) use a quantum LSTM model to train a language model, which is then transferred to classical devices for sentiment analysis. Their work conducts various experiments, adjusting parameters such as the number of qubits and local epochs, and achieves results that are comparable, if not superior, to those of the classical

counterpart. Building on the discussion of articles focused on natural language processing, Yang et al. (2022) introduce a VQC into their Bidirectional Encoder Representations from Transformers (BERT) model and their vertical federated architecture to demonstrate the competitive performance on text classification. Along similar lines, Qu et al. (2024) propose a QFL framework using a quantum recurrent neural network for speech emotion recognition. Through experimental simulations and analysis, the study demonstrates that the proposed algorithm outperforms existing quantum models, such as quantum LSTM, achieving higher recognition accuracy and faster convergence rates during training.

Last but not least, the studies by Du et al. (2022) and Huang et al. (2022) not only perform classification tasks on the MNIST dataset but also apply a QFL framework to find the ground state of a chemical molecule. Additionally, the latter work extends their framework to a quantum approximate optimization algorithms (QAOA) to successfully tackle the well-known Max-Cut problem.

6.1.2 Trends and challenges regarding efficiency

Benchmarking performance As discussed in the previous section, many of the works that are not review-based focus on enhancing the performance of the proposed frameworks, with 53 out of 85 papers dedicated (sometimes not exclusively) to this goal. Furthermore, as previously mentioned, 48 of these 53 papers focus on performing a classification task. Twenty-two of these concentrate on binary classification, while 23 make use of the MNIST dataset.

These works typically compare their proposed models either to a centralized quantum or classical version or to a fully federated classical model. Given this, many articles show that within a QFL scheme, they can achieve similar performance to centralized models, concluding that QFL provides comparable accuracy while gaining the benefits of decentralization, such as enhanced privacy. However, despite the proliferation of works that use quantum models and resources, there is no clear justification for their use in these contexts. The tasks addressed often do not present challenges that inherently require quantum solutions, leaving open the question of why quantum techniques are necessary to tackle these problems when (federated) classical methods suffice (Schuld and Killoran 2022).

Furthermore, binary classification tasks, particularly those using datasets like MNIST, may be relatively straightforward for classical computational methods. Although some of the works reviewed suggest that quantum frameworks may offer marginal improvements, a proper benchmarking process is often lacking. For example, works such as the foundational article by Chen and Yoo (2021) or the ones from Song et al. (2024); Ma et al. (2024); Rofougaran et al. (2024); Qu et al.

(2024), to name a few, use classical preprocessing techniques like principal component analysis to reduce dimensionality and address the current limitations of quantum resources, changing the nature of the problem being tackled. As a result, accurately benchmarking the performance of quantum models on these tasks, particularly when using widely established machine learning datasets, becomes both complex and problematic (Bowles et al. 2024).

Trainability and Barren plateaus On another note, the concept of barren plateaus is becoming increasingly important in the field of QML. Barren plateaus are a significant challenge in training VQC, where the gradient of the loss function becomes exponentially small as the system size increases. This issue arises during optimization, making it extremely difficult for gradient-based methods to update the parameters effectively. As stated in Larocca et al. (2024), selecting an inappropriate quantum model configuration (i.e., ansatz), initial state, measurement, loss function, or accounting for hardware noise can lead to barren plateau issues, preventing the model from learning. Thus, addressing barren plateaus is crucial for the scalability of quantum models.

Nonetheless, it is worth noting that in broader quantum computing research, barren plateaus are not always purely detrimental: for instance, Gong et al. (2024) show they can be exploited for enhancing robustness against adversarial attacks, while Wang et al. (2025) explore successful mitigation techniques in many-body quantum systems, suggesting that such strategies might eventually benefit QFL frameworks as well.

QCNN are among the most prevalent quantum models that do not suffer from the barren plateau problem (Larocca et al. 2024). In QFL, 4 studies adopt this approach; however, none offers a justification for this choice concerning the issue of barren plateaus. In fact, only 2 of all reviewed articles reference this concept: on the one hand, Qiao et al. (2025) identifies barren plateaus as one of the two primary challenges in QFL (the other being the impact of quantum noise on model performance); on the other hand, Wu et al. (2024) also identifies the issue of barren plateaus and briefly discusses possible solutions, such as careful parameter initialization or problem-specific ansatzes.

As discussed in Cerezo et al. (2024), barren plateaus tend to arise when working within exponentially large Hilbert spaces. Approaches that mitigate this issue, such as QCNNs, do so by employing localized operations that significantly reduce the dimensionality of the problem. However, this characteristic seems to imply that these approaches could be effectively classically simulable (Bermejo et al. 2024) by techniques such as tensor networks and shadow tomography. Therefore, as we will elaborate on in the next section, we believe that this emerging research area should be considered in the development of QFL frameworks.

A few studies explore related topics in this area. Chehimi et al. (2024a) is the only work to propose that bounding the trainability of quantum models within the QFL framework, such as by incorporating the quantum Fisher Information metric, represents a promising research direction. Ma et al. (2024) is the only study that compares the expressivity of their quantum model with that of its classical counterpart. Finally, Song et al. (2024) introduce a QFL framework that employs shadow tomography, a technique that eliminates the need for clients to utilize quantum resources.

Data and system heterogeneity A key discussion within the FL community concerns the fact that real-world data is often non-independent and identical distributed (IID), which can significantly impact model performance (Guendouzi et al. 2023). Of all the articles reviewed in this category, only 12 acknowledged or addressed this issue in their experiments, showing that quantum models also experience performance degradation in non-IID data settings and proposing different methods to tackle this issue (Gurung et al. 2023a; Park et al. 2024; Bhatia and Neira 2024; Zhao 2023; Yamany et al. 2023; Gupta et al. 2024; Kais et al. 2023; Yun et al. 2022; Chehimi and Saad 2022; Qu et al. 2024; Xia and Li 2021; Shi et al. 2024).

Other significant discussion areas within the FL community are model and system heterogeneity (Saha et al. 2024). These concepts are related, but while the former refers to using different machine learning models on a FL setting, the latter pertains to the variations in power and capabilities among the devices involved. Only a couple of studies explore one of these concepts to some extent (Singh et al. 2023; Sünkel et al. 2024).

In this regard, it is also relevant to point out that only the studies by Rayhan (2023) and Chehimi and Saad (2022) involve experimentation with quantum data, that is, data obtained from some quantum mechanical process within a (simulated) quantum device.

Current quantum hardware Along these lines, the decentralized scheme of QFL may appear to be advantageous in today's context, where NISQ devices are prevalent, as noted in Section 3. By distributing computation, QFL could help mitigate the limitations of individual quantum devices, such as limited qubit counts and high noise levels. However, only 7 articles take noise into account in their analyses (Du et al. 2022; Rofougaran et al. 2024; Innan et al. 2024; Sahu and Gupta 2024; Innan et al. 2024; Qu et al. 2024; Xia and Li 2021).

On top of that, only the studies by Innan et al. (2024) and Du et al. (2022) appear to conduct experiments on actual quantum hardware, both utilizing IBM's quantum chip.

As discussed in Section 6.1.3, further research in this area is needed.

6.1.3 Future directions regarding efficiency

Beyond classification tasks While most of the studies in QFL have focused primarily on classification tasks, it is crucial to explore a wider range of applications to fully harness the potential of QFL. Beyond classification, areas such as optimization, time-series analysis, and complex decision-making problems offer promising directions for research. Expanding QFL applications into these areas could reveal the unique advantages of quantum computing and foster innovation in both the quantum and federated learning domains.

Moreover, a key challenge in QFL is the lack of a clear justification for using quantum resources, especially when the problems being tackled, such as binary classification or tasks using MNIST, can already be solved efficiently by classical methods. Thus, future research could focus on developing more suitable benchmarks that reflect the strengths of quantum systems or explore more complex tasks where quantum solutions offer a distinct advantage within the FL setting.

Trainability and expressivity Investigating whether a QFL framework could enhance the trainability or expressivity of quantum models presents a highly promising avenue of research. Given the challenges posed by barren plateaus, where gradients vanish during training in high-dimensional quantum systems, QFL could potentially offer a collaborative learning environment that mitigates these issues by leveraging diverse data distributions or model updates across clients. Additionally, exploring whether QFL can introduce new optimization techniques or learning paradigms that improve quantum models' ability to converge faster and generalize better is another key area worth pursuing. This line of research may reveal new insights into how decentralized quantum learning can address fundamental challenges in quantum computing and make quantum models more practical for a broader range of applications.

Aggregating different quantum models or devices It would be interesting to explore whether completely different quantum models or distinct quantum devices could learn from one another in a FL framework. By facilitating collaboration among diverse quantum systems, researchers could uncover innovative strategies for leveraging the unique strengths of each model or device, ultimately advancing the capabilities of quantum machine learning and fostering more robust solutions to complex problems.

Realistic implementation Introducing noise into experiments is essential for advancing research in QFL, especially in the current NISQ era. By incorporating noise into simulations, researchers can evaluate whether the federated scheme is more resilient to noise compared to centralized models. While a few studies have begun to explore this topic, for

instance, by adding depolarizing or dephasing noise to local computations, a broader and more systematic investigation is still lacking. Key open questions include how different types of quantum noise affect local training processes and aggregation steps, and whether federated setups can naturally isolate or contain errors. These would provide valuable insights into QFL's potential to manage noise-related challenges, such as error mitigation, and demonstrate its advantages in real-world quantum computing scenarios.

In addition, it is essential to experiment on real quantum hardware to assess the feasibility of deploying QFL frameworks in the near future. Simulations, while valuable, cannot fully capture the complexities and challenges presented by real-world quantum devices, such as noise, error rates, and hardware limitations. By conducting experiments on actual quantum processors, researchers can better understand how QFL performs in practice and address potential obstacles, paving the way for more reliable and scalable quantum solutions in decentralized learning environments.

Harnessing quantum data Only a few articles focus on quantum data, revealing a significant gap in the current research landscape. Most studies rely on classical datasets, leaving the potential of quantum-generated data, such as that from quantum sensors or other quantum devices, largely unexplored. This presents a valuable opportunity for future research to investigate how quantum data can be effectively integrated into QFL frameworks. By leveraging quantum-derived datasets, researchers could unlock new applications where quantum processing offers distinct advantages, moving beyond traditional approaches and harnessing the unique properties of quantum systems.

6.2 Security

6.2.1 Baseline and literature review

Out of the 103 articles reviewed, 27 focus to some extent on preserving privacy or attempting to mitigate potential threats and data leaks.

As reviewed in the previous section, one of the first articles to introduce QFL (Chen and Yoo 2021) primarily focuses on enhancing efficiency. However, the authors highlight, as future work, that updated and global parameters that are exchanged between clients and the server could be vulnerable to attacks, allowing malicious entities to extract sensitive information about clients' local data.

Building upon this idea, the work by Li et al. (2021) provides an experimental demonstration of such a vulnerability through a specific type of attack known as the gradient attack. In this scenario, adversaries can infer local data by analyzing the gradients shared during the training process. To

counteract this threat, the authors apply DP to the exchanged parameters, adding carefully calibrated noise. Their experimental results demonstrate that this approach effectively reduces the risk of gradient attacks in the QFL setting, thereby enhancing the security of client data. Similarly, Chu et al. (2023) present a QFL framework in which the authors use HE to encrypt the parameters before sending them to the central server.

However, Li et al. (2021)'s most significant contribution lies in the use of blind quantum computing (BQC), which enables the delegation of local quantum computations to an external quantum server while ensuring that no information is disclosed (Broadbent et al. 2009). In this way, unlike Chen and Yoo (2021), where clients perform local quantum computations, this framework only requires clients to prepare and transmit single qubits. This approach could be more realistic, where, for instance, several hospitals could afford the quantum computational costs and would highly benefit from the FL architecture.

In the BQC context, Polacchi et al. (2024) emphasize that security should encompass not only protecting data and computations but also ensuring that the server actually performs the requested computations. To address this, the authors present an experimental implementation of a two-client verifiable BQC protocol within a distributed architecture. Along similar lines, Drmota et al. (2024) demonstrate a verifiable BQC protocol using trapped ions and single photons, highlighting a feasible path toward secure quantum cloud computing with physically separated clients and servers.

Another approach to enhance security in QFL leverages the inherent properties of quantum mechanics. For instance, the work from Zhang et al. (2023) focuses on developing a secure parameter aggregation using entangled qubits. This article examines the efficiency and security of this protocol in a FL architecture, both with a server and in its absence, where a random client then assumes the server's role in each iteration. In contrast to HE or DP, this secure aggregation procedure does not decrease the model's efficiency. However, it relies on teleporting entangled states over long distances, which presents a significant limitation due to the constraints of current hardware, as will be discussed later. Similar approaches and results are presented in Li et al. (2024).

Additionally, Chen et al. (2024) stand out as a unique contribution to the field. The authors conduct extensive numerical simulations to illustrate that incorporating noise into quantum computations enhances the robustness of the models against specific attacks, all while maintaining a high level of accuracy.

The work by Kumar et al. (2023) also leverages the intrinsic properties of QML models to safeguard against data leakage in FL. Building on the findings of Schuld et al. (2021), the authors experimentally demonstrate that using highly

expressive quantum embedding maps for classical data provides inherent protection against gradient inversion attacks, effectively enhancing privacy within the FL framework.

An alternative method for enhancing privacy in QFL using quantum mechanics involves utilizing QKD. In a nutshell, QKD is a cryptographic technique that uses the principles of quantum mechanics to securely distribute encryption keys between parties, ensuring that any eavesdropping attempts are detectable due to the disturbance caused by measuring quantum states. This technology provides a higher level of security compared to classical encryption methods, as any interception of the quantum key would be immediately noticeable (Bennett and Brassard 2014; Ekert 1991). The works by Kaewpuang et al. (2022), Namakshenas et al. (2024), Park and Lee (2023), Xu et al. (2023a), Ren et al. (2024), Xu et al. (2023b), Xu et al. (2022), and Ren et al. (2024) center around the implementation of a QKD protocol.

Finally, the PhD thesis by Xia (2021) builds on the author's earlier works, such as Xia and Li (2021); Xia et al. (2021), where Byzantine issues in distributed machine learning, and specifically in QFL, are explored. These attacks occur when certain workers intentionally send incorrect parameter updates to the server during the training phase. In the thesis, the author first introduces a QFL framework utilizing QNNs, consistent with most works in Section 6.1. Byzantine attacks are then adapted to their quantum equivalents, discussing their applicability in the proposed QFL framework and experimentally proposing potential strategies for mitigating these threats when necessary.

6.2.2 Trends and challenges regarding security

Quantum communications A strong trend within the articles focusing on enhancing privacy and security in QFL involves the simulation of quantum communication techniques, such as the use of entangled-pair particles or QKD protocols. In fact, 10 out of the 27 articles considered in this section use one of these techniques within communications in their studies. These methods are seen as natural extensions of quantum technology because they offer privacy and security levels that classical communication protocols cannot match. The intrinsic properties of quantum mechanics, such as the no-cloning theorem and the ability to detect eavesdropping through measurement disturbances, make quantum communication a powerful tool for securely exchanging information in a QFL framework.

However, the practical deployment of such quantum communication methods faces a critical challenge: the current quantum hardware required to implement these protocols is still in its early stages of development. The creation and manipulation of entangled particles over long distances or the reliable implementation of QKD in real-world environments require sophisticated and costly equipment that is not

yet widely available. Moreover, building and scaling such quantum communication networks to handle the complexity and scale of federated learning in diverse industries presents another significant obstacle. These technological hurdles limit the immediate applicability of quantum communication in QFL beyond simulations and controlled experimental setups (Larasati et al. 2022).

Furthermore, despite the promise of these techniques, it is important to recognize that all existing works in this area rely on ideal simulations, which assume perfect quantum operations and no external interference. In real-world conditions, these assumptions may not hold. For example, side-channel attacks, where an adversary exploits indirect information leaked during the execution of the real implementation of a quantum protocol, could still be possible.

Classical plugins There also exists an emerging trend in QFL research where classical encryption and security techniques, such as DP and HE, are integrated into quantum frameworks (Ren et al. 2024; Dutta et al. 2024; Rofougaran et al. 2024; Li et al. 2021; Li and Deng 2024). Another example is the work by Gurung et al. (2023a), which introduces a novel approach by combining QFL with blockchain technology. While exploring the integration of these classical methods in QFL is valuable, we think that they do not necessarily enhance or justify the use of quantum resources. Additionally, a significant challenge is ensuring that the introduction of these techniques does not degrade the model's performance or increase communication costs exponentially.

Another common approach is combining FL with post-quantum cryptography methods, as seen in works like Li et al. (2024) or Zhang et al. (2024). However, these studies are not included in this survey, as they rely exclusively on classical models and resources rather than incorporating quantum technologies.

Mitigating attacks Seven articles within the reviewed literature focus specifically on mitigating attacks such as gradient inversion or Byzantine attacks, among others. These works aim to protect the integrity and privacy of data in a QFL setting by addressing threats that occur during the communication and model update phases. For example, gradient inversion attacks attempt to reconstruct client data from the gradients shared with the central server, while Byzantine attacks involve malicious clients deliberately sending corrupted updates to disrupt the training process. While it is certainly valuable to explore how such attacks manifest within the QFL context, these challenges are also present in classical FL as well.

Although studying these attacks in QFL contributes to our understanding of potential vulnerabilities, most of these works rely solely on classical mitigation strategies, without fully leveraging the advantages of quantum technology.

The challenge, then, lies in developing mitigation protocols that not only address security threats like gradient inversion or Byzantine attacks but also justify the use of quantum resources. To fully realize the potential of QFL, these solutions must leverage unique quantum properties, ensuring that quantum technology offers clear advantages over classical methods in protecting data and ensuring secure learning against specific attacks.

6.2.3 Future directions regarding security

Private and trustworthy delegated quantum computations With the current limitations of quantum hardware, exploring innovative solutions to delegate quantum computations, such as BQC, within a FL framework offers a promising research direction. As quantum devices are still in their early stages and not widely accessible, most potential users lack the resources to run advanced quantum computations locally. This limitation becomes especially relevant in QFL, where multiple clients are expected to perform quantum computations and share updates with a central server. By incorporating delegated quantum computations without revealing any information into QFL, it would be possible to offload the quantum computational tasks to more capable quantum servers, while ensuring that the clients maintain the privacy of their data and computations.

Additionally, as highlighted by Polacchi et al. (2023), ensuring that the delegated computations are performed correctly is crucial and emerges as a promising research direction in the context of QFL. Verifiable quantum computing would ensure that computations performed on remote quantum servers are both accurate and trustworthy, providing clients with verifiable proof of correctness.

Similarly, protocols like multiparty summation and related techniques could be adapted in a QFL setting to securely aggregate data from multiple clients while maintaining privacy.

By integrating these advanced techniques, researchers could address potential vulnerabilities and enhance the reliability of QFL, making it more practical and secure. This exploration could also facilitate the adoption of quantum technologies in real-world industries, even in the near term, where quantum hardware capabilities remain limited.

Inherent privacy-friendly implementation One of the most promising directions in enhancing privacy and robustness in QFL lies in developing models that are inherently more secure and resistant to attacks. For instance, as mentioned in the previous section, the work by Chen et al. (2024) introduces noise directly into the quantum computations, demonstrating that this method increases the model's robustness against certain types of attacks without significantly affecting its accuracy. By incorporating noise strategically,

they provide a built-in layer of defense within the model itself, ensuring that privacy is preserved even under potential threats.

Similarly, the work by Kumar et al. (2023) explores the use of various quantum embeddings to naturally protect against gradient inversion attacks. By leveraging highly expressive quantum maps to embed classical data into quantum states, the authors show that certain quantum features can inherently provide a level of privacy.

These approaches suggest that, rather than relying on external techniques, quantum models can be designed with built-in defenses that make them more secure.

Thus, it is crucial to explore how quantum-native techniques can be used to enhance security or mitigate attacks. Moreover, incorporating quantum-specific solutions, such as using inherent quantum properties within a quantum model, offers a more compelling reason to adopt QFL, providing stronger justification for the integration of quantum resources in tackling privacy and security challenges within this field.

Quantum secure direct communications QSDC represents an intriguing field of exploration within the context of QFL. While QKD focuses on securely distributing encryption keys over quantum channels, QSDC allows for the direct transmission of confidential information between parties without the need for prior key exchange. This characteristic makes QSDC a powerful tool in QFL, where multiple decentralized nodes collaboratively train machine learning models without sharing raw data. Exploring QSDC as part of the QFL framework can open up new avenues for safeguarding communication between distributed nodes while capitalizing on the inherent security benefits of quantum mechanics. By integrating QSDC, QFL could become even more robust and resistant to eavesdropping or other security threats, making it a compelling area for further research and development.

Hardware advancements and real quantum implementation Implementing QKD safely within a QFL framework holds significant promise for achieving unparalleled levels of privacy and security. QKD's ability to leverage the principles of quantum mechanics to detect eavesdropping provides a clear advantage over classical encryption methods. However, despite its potential, implementing QKD in practice is far from straightforward. Current quantum hardware remains limited in terms of scalability, and the practical deployment of QKD protocols is still in its infancy. Additionally, even with theoretically secure quantum communication, side-channel attacks, where vulnerabilities in the physical implementation are exploited, remain a serious concern.

Looking further ahead, the ability to deploy quantum entanglement across multiple nodes could represent one of the most transformative developments for QFL. If clients and servers could communicate through entangled quan-

tum states, it would significantly mitigate key challenges present in classical FL, particularly communication latency and privacy leakage. Such entanglement-enabled communication could eliminate the need to explicitly transmit sensitive model updates, enabling intrinsically private and highly efficient learning schemes.

To fully understand the viability of QKD within QFL, it would be critical to implement these protocols on real quantum hardware rather than relying solely on simulations. Real-world tests would allow researchers to assess the practicality, scalability, and robustness of QKD in the FL context. This would also provide an opportunity to identify potential side-channel attacks specific to QFL implementations. By addressing these challenges experimentally, the field could move closer to practical, scalable solutions that integrate quantum communications with secure machine learning architectures.

6.3 Implementation

6.3.1 Baseline and literature review

In this section, we explore the body of work focused on implementing and deploying QFL in practical scenarios. These articles delve into real-world applications, adapt the QFL framework for specific use cases, or propose modifications to make the frameworks more feasible and scalable in current technological landscapes. By addressing the challenges of integrating quantum technologies with federated learning, these studies take significant steps toward bridging the gap between theoretical advancements and tangible implementations. Among the various domains explored, healthcare stands out as a primary application area of QFL, due to the inherently distributed and privacy-sensitive nature of medical data. Because of its unique relevance and recurring presence in the literature, we discuss QFL applications in healthcare separately in Section 6.4. Other application scenarios, such as autonomous vehicles and satellite-based systems, are covered in the present section. Out of all the articles reviewed, we identified 31 that fall to some extent within this section.

One of the early works that explores a real-world application of QFL is the study by Narottama and Shiny (2020), which constructs a two-layer network using QNNs within a wireless communication architecture. In a follow-up study, Narottama et al. (2021) further optimized the QFL framework by reducing the computational and communication burdens on clients. This was achieved by allowing clients to send only statistical summaries of their data rather than full datasets. This innovation proved more efficient in the context of non-orthogonal multiple access (NOMA) power allocation for wireless networks. NOMA is a wireless communication technique that allows multiple users to share the

same frequency channel by assigning different power levels to their signals, thereby improving spectral efficiency and enabling more users to be served simultaneously. Also, the work by the same authors (Jamaluddin et al. 2023) investigates this setting for unsupervised learning. Their research continued in Narottama and Shin (2023) and Narottama et al. (2024), in which the authors added quantum teleportation as a method for resource optimization in future wireless communication systems.

There exist more articles within the literature that explore the intersection of QFL and wireless communications, aiming to improve communication efficiency and adaptability in complex network environments. For instance, Elfaki et al. (2023) focus on optimizing communication by reducing data transmission sizes, thereby enhancing robustness in unstable networks, which is essential for efficient QFL implementation.

Building on this theme, Yun et al. (2022) propose a new QFL framework, dubbed eSQFL, which aims to make the communication process more efficient while adapting to varying conditions in wireless channels and computing resources. Their approach demonstrates promising results, although it focuses only on a simple image classification task.

Moreover, Zaman et al. (2023) explore the potential of QFL and QAOA to meet the stringent requirements of ultra-reliable and low-latency communication (URLLC) in 6G networks. Their work addresses critical challenges such as wireless scheduling and vehicle routing, showcasing how quantum technologies could advance communication infrastructures in these cutting-edge scenarios.

On the other hand, several works in the literature focus on optimizing client selection, server selection, and resource allocation.

For example, one particularly innovative study by Wei et al. (2024) stands out as the only work to employ quantum annealers from D-Wave for optimizing client selection in QFL. This approach introduces practical applications of quantum technologies into the client selection process, offering a novel way to leverage quantum computing for real-world federated learning tasks. By using quantum annealers, the study opens up new pathways for client selection optimization that could improve the scalability and efficiency of QFL systems.

Gurung et al. (2023c) propose a dynamic server selection model combined with a signature protocol for parameter updates in QFL. Their work demonstrates the robustness of the proposed model through numerical analysis, showcasing how flexible server selection can enhance system performance and security.

Resource allocation and cost optimization also receive considerable attention in QFL studies. For example, Xu et al. (2022) and Kaewpuang et al. (2022) focus on optimizing resource allocation and routing for FL workers. Their

research aims to minimize the deployment costs of QKD nodes, highlighting the importance of resource management in the context of secure quantum communications, as discussed in previous sections.

Another notable work by Chaudhary et al. (2024) merges QFL with reinforcement learning to tackle mode selection and resource allocation in a vehicle road cooperation system scenario. Through extensive numerical analysis, this study demonstrates how reinforcement learning can be effectively combined with QFL to optimize both computational resources and communication efficiency in real-time systems, further pushing the boundaries of QFL's applicability in complex environments.

This article is not the only one focusing on autonomous vehicles; in fact, several other studies have explored the integration of QFL in this area as well.

One such contribution comes from Xu et al. (2023b), who propose a QKD protocol explicitly tailored for quantum autonomous vehicular networks in a federated learning environment. This research highlights the importance of secure communication channels for autonomous vehicles, ensuring that sensitive data shared among vehicles remains confidential and protected from external attacks.

Building upon the need for robust security measures, Yamany et al. (2023) introduce a QFL model as part of a larger cyber defense framework aimed at protecting autonomous vehicles from adversarial attacks. Their focus is on developing security mechanisms that can defend against increasingly sophisticated threats, enhancing the safety and reliability of autonomous vehicle networks operating in real-world scenarios. By leveraging quantum technologies, their approach strengthens the overall resilience of these systems against external manipulation.

Similarly, Kim (2023) also explores the use of QFL in the context of autonomous vehicles, but with a specific emphasis on applying it to realistic, real-world scenarios. Although their work does not include numerical experiments, it suggests the potential of quantum-enhanced learning models to tackle challenges faced by autonomous vehicles, such as decision-making in complex environments.

Finally, Hazarika et al. (2024) address the challenges of a vehicular metaverse by offering a cost-efficient and adaptive solution for managing dynamic vehicular landscapes within a QFL framework. Their research introduces a novel approach to optimize communication and computational resources for vehicles operating in constantly changing environments, underscoring the role of QFL in shaping the future of autonomous transportation systems.

In addition to autonomous vehicles, several studies have also explored applications involving satellites. For instance, Park et al. (2024) emphasize satellite-ground communications and presents a QFL framework that demonstrates significant computational and communication efficiency compared

to traditional methods. As already mentioned in Section 6.1, this is one of the few articles that consider non-IID data within their experiments. Similarly, Wang et al. (2024) investigate space-air-ground integrated networks, where satellites and aerial vehicles, such as drones, collaborate with quantum computers within a QFL framework. This approach aims to train classification models while ensuring secure and reliable communication transmissions, highlighting the potential of quantum technologies in enhancing satellite communication systems.

Last but not least, a novel approach proposed by Daskin (2024) introduces the use of quantum chips on client devices, which operate based on an input quantum state. In this setup, the server sends a quantum state to the clients; this state represents a superposition of the model's output with slightly altered parameters. Using these superpositioned states, each client calculates its local gradient and sends it back to the server. Since the quantum chips are controlled by the quantum states themselves, clients do not need to have direct knowledge of the model.

These diverse approaches illustrate how QFL is being adapted to meet the demands of real-world use cases while addressing key challenges like scalability, security, and communication efficiency.

6.3.2 Trends and challenges regarding implementation

Wireless and poor channel communication As presented in the previous section, many articles concentrate on implementing QFL in environments characterized by wireless communications or poor channel conditions. While improving communication efficiency in such scenarios is an important consideration, we believe that this is an issue that also arises in classical FL. More importantly, these articles do not address communication inefficiencies and unreliable channels using novel quantum approaches; rather, they often implement quantum computing to accelerate certain subroutines or computations more efficiently without fully leveraging the unique potential of quantum technologies (Narottama and Shiny 2020). In some cases, the models used are entirely classical, with quantum resources applied in a completely different task (Kaewpuang et al. 2022). This approach does not provide a compelling justification for the integration of quantum resources, as the focus remains on classical challenges rather than the exploration of quantum-specific advantages that could enhance the overall QFL framework.

Client selection and resource allocation Another notable trend in the literature is the use of quantum computing to efficiently optimize client selection during the training phase of QFL. This selection is typically based on factors such

as local model performance, the quality of local data, or communication overhead. While this optimization serves as just one subroutine within the broader QFL framework and may not have immediate implications for near-term quantum federated learning implementations, it is still a highly interesting approach. This is because it opens the door to incorporating other quantum models, such as the QAOA or quantum annealing, to perform approximate optimization tasks at a faster rate than classical methods. Notably, the works by Wei et al. (2024) and Huang et al. (2022) are the only ones in the current body of literature to explore the use of these quantum models in QFL.

Real-world applications As discussed in the previous section, a significant number of works in the field of QFL focus on real-world applications. A prominent area of interest is the development of QFL models for autonomous vehicles, where researchers explore the integration of quantum technologies into vehicular networks to enhance security and performance. This trend reflects the potential of quantum computing to address the growing demands of such complex systems. Additionally, some studies extend beyond autonomous vehicles, delving into other emerging domains like the Metaverse and satellite communications, where QFL frameworks are proposed to improve communication efficiency and security.

However, despite the promising nature of these works, they generally focus on tailoring a general QFL framework to specific applications rather than testing these models on actual quantum hardware. While these studies provide valuable theoretical insights and simulation-based results, they often rely on idealized quantum environments. Consequently, the challenge of implementing real QFL models on existing quantum devices remains largely unexplored.

6.3.3 Future directions regarding implementation

Delegated quantum computations As mentioned in Section 6.2, delegated quantum computation protocols, such as BQC, could hold significant promise in the realm of QFL. While we previously discussed the importance of delegating quantum computations securely (ensuring that no information is revealed) and in a verifiable manner (allowing for the validation of correctly executed delegated computations), we now wish to highlight their potential impact in terms of resource efficiency. Given the limited and often costly quantum resources available today, the exploration of these delegated computation protocols may lead to the development of application-specific and economically viable QFL frameworks. By leveraging such techniques, it becomes possible to optimize the use of available quantum resources, thereby making QFL more practical and scalable in real-world scenarios.

Emerging role of hybrid quantum algorithms Hybrid quantum-classical algorithms such as QAOA, variational quantum eigensolver (QVE), and other variational approaches offer a promising path forward for integrating quantum capabilities into FL. These methods are particularly attractive due to their compatibility with NISQ devices and their potential for efficient optimization, which is often central to QFL tasks such as model coordination, client selection, or resource scheduling. While their current use in the literature is still limited and typically restricted to auxiliary subroutines rather than full end-to-end training, these algorithms open the door to incorporating more adaptable and scalable quantum components. As the hardware and software ecosystems continue to mature, we anticipate that hybrid algorithms will play a more central role in enabling practical and efficient QFL deployments, especially in settings where communication efficiency and optimization are critical.

Quantum-inspired algorithms Along similar lines, quantum-inspired algorithms present an intriguing avenue for exploration within the context of QFL. These algorithms can often be implemented (almost) immediately and with relative ease on existing classical hardware, making them accessible for practical applications.

Notably, tensor networks have been explored for distributed learning tasks, providing compact model representations that reduce communication costs and improve scalability in federated environments (Bhatia et al. 2025). Similarly, quantum-inspired techniques for singular value decomposition and dimensionality reduction have been proposed to optimize data transmission and local model efficiency and could be beneficial in FL settings (Gily'en et al. 2018; Jethwani et al. 2020; Yao et al. 2022).

By utilizing these techniques, researchers can effectively bridge the gap between classical and quantum paradigms, facilitating the integration of quantum concepts into fFL frameworks without the need for extensive quantum infrastructure. This could lead to innovative solutions that harness the principles of quantum computing while remaining grounded in practical implementation, ultimately paving the way for more widespread adoption of QFL.

Quantum-specific advantages Rather than concentrating solely on optimizing efficient subroutines with quantum computing to facilitate QFL in wireless or low-quality communication environments, we believe that one of the primary focuses should be on exploring whether quantum computing provides inherent advantages in communication protocols. This is closely tied to concepts such as QKD and other security protocols that may inherently offer greater security than classical communication methods.

In the context of QFL, where communication rounds are frequent and data privacy is paramount, the theoretical

advantages of quantum communication protocols, particularly QKD, offer meaningful benefits over classical approaches. QKD provides information-theoretic security, allowing for the detection of eavesdropping and ensuring secure parameter exchange even against adversaries with unlimited computational resources. Additionally, exploring the use of quantum entanglement and teleportation to reduce communication complexity or latency represents a promising research direction, though such approaches remain largely theoretical and contingent on future advancements in quantum networking infrastructure.

Beyond just enhancing security, it would be valuable to investigate whether quantum communications can also prove to be more practical, scalable, and accessible in various contexts.

Real hardware implementation We believe that a more important and realistic near-term focus, rather than addressing challenges related to, for instance, changing wireless conditions, is the actual implementation of real quantum computers within a general QFL framework. As mentioned in previous sections, only a limited number of studies have successfully implemented QFL on actual quantum machines. Prioritizing the integration of functional quantum hardware will pave the way for more meaningful advancements in QFL, showcasing the genuine benefits of quantum resources.

We believe that QFL represents a promising opportunity to leverage quantum models, especially given its distributed learning framework within the current NISQ era. Thus, while the focus on domain-specific applications is important, the next step should involve practical testing and real-world deployment of QFL on quantum hardware to truly demonstrate its potential and feasibility.

In addition, real-world scenarios often come with specific characteristics and settings that impact the performance of models. For instance, the presence of non-IID data among clients or the introduction of different kinds of noise is crucial, as it can greatly affect the overall performance of a model, as highlighted in previous sections. Notably, only the works by Zhao (2023); Yamany et al. (2023); Park et al. (2024); Qu et al. (2024) take non-IID data into consideration, while only the works by Sahu and Gupta (2024); Qu et al. (2024) introduce some kind of noise into their experiments. Therefore, these concepts present a significant opportunity for future research.

6.4 Healthcare

In this subsection, we focus specifically on the application of QFL in the healthcare sector. As commented previously in this work, the healthcare domain has greatly benefited from classical FL, as medical data is critical for improving

diagnostic accuracy but also highly sensitive and subject to strict privacy regulations.

Among the articles addressing healthcare applications, only the one by Park and Lee (2023) specifically focuses on enhancing security through the use of QFL. In this work, the authors leverage a QKD protocol and a FL scheme to successfully predict COVID-19 and pneumonia through classification tasks while ensuring the preservation of local privacy.

The majority of the other studies focus on developing general QFL models, much like those described in the efficiency category, using them to classify various healthcare datasets. For instance, Hu et al. (2021) leverages a VQC and a general FL structure to enhance brain-computer interface systems, particularly in classifying electroencephalogram data for motor imagery tasks. Similarly, Innan et al. (2024) focus on classifying breast cancer datasets, while De et al. (2024) perform early diabetic retinopathy detection using faster convergence times and fewer samples in their simulations.

Analogously, Qu et al. (2023) propose a digital-twin framework using QFL for intelligent diagnosis in 5G networks, with experiments on healthcare datasets showing promising results, though they acknowledge a performance gap compared to classical methods, likely due to reliance on simulators rather than actual quantum hardware.

Notably, Bhatia et al. have made significant contributions to the field. In Bhatia et al. (2023a), they applied QFL techniques to various medical datasets, including medical scans and COVID-19 records, introducing optimizations to reduce the computational and communication load on clients. Building on this, Bhatia et al. (2023b) and Bhatia et al. (2024) developed a QFL framework specifically for medical image classification, demonstrating improved generalization on non-IID datasets, even when faced with unbalanced and insufficient training data. Most recently, in Bhatia and Neira (2024), they employed tensor networks to simulate quantum circuits for the classification of more medical datasets, such as chest X-rays and brain and kidney scans, further advancing their work in medical applications of QFL.

Remarkably, the use of non-IID data is crucial in this domain, as healthcare experiments aim to address real-world classification tasks that are expected to emerge in the near future. Moreover, medical data is frequently non-IID, reflecting the diverse and heterogeneous nature of patient information. Along these lines, Qu et al. (2024) applies fuzzy logic to perform extended experiments on the COVID-19 dataset, exploring the model's performance on non-IID data and under noisy conditions. Similarly, the work by Gupta et al. (2024) addresses mental state classification using the wearable stress and affect detection dataset and also focuses on non-IID data.

Lastly, a unique contribution comes from the work by Sahu and Gupta (2024), where the authors simulate noisy

quantum devices to evaluate the performance of a quantum classifier on the chest X-ray and pneumoniaMNIST datasets. In this study, they explore the impact of realistic noise in quantum hardware, which is crucial for understanding the challenges of deploying QFL on near-term quantum devices. Their findings provide valuable insights into the limitations and potential of nNISQ devices for medical image classification, shedding light on the need for more robust algorithms that can withstand the imperfections of current quantum hardware.

While, in general, these models are effective in leveraging QFL for medical diagnosis and prediction, their primary focus is on improving performance and efficiency rather than addressing the unique security challenges inherent to handling sensitive healthcare information. Furthermore, none of these experiments were conducted on real quantum hardware, which underscores the gap between advancements in simulations and their practical implementation in this field.

7 Concluding remarks

In this paper, we have reviewed the state-of-the-art of QFL. We explicitly conducted a thorough and systematic search to identify relevant articles, an effort that, to the best of our knowledge, had not been previously undertaken. In addition, we have proposed a novel taxonomy to categorize articles based on their primary focus and conducted several analyses on the selected articles to provide additional characteristics. Thus, we believe this review can help readers picture a general map of the topic's current state, providing readers with a clear and general overview and enabling them to identify and engage more directly with research that aligns closely with their specific interests.

While previous sections explored category-specific findings, this final section synthesizes the key insights from our analysis, emphasizing overarching themes that span all categories and highlighting specific challenges that we believe are essential for advancing research in particular areas.

All in all, the examination of current trends, along with the proposed challenges and future directions, seeks to work towards a QFL model that is (i) compatible with NISQ devices, (ii) demonstrates a quantum advantage, and (iii) has practical applications.

Deployment in real hardware Implementing QFL on real hardware is crucial for evaluating its practical applications and potential benefits in the near future. Currently, the vast majority of research in this area relies on simulators rather than actual quantum devices, which limits our understanding of QFL's performance in realistic environments, such as those using current NISQ devices. Many aspects could be explored in this context; for instance, what happens when we

introduce hardware noise? How significantly does it impact performance? Can the distributed nature of QFL enhance the effectiveness of error mitigation strategies? Does the inherent quantum noise contribute to improved model security?

Additionally, examining the feasibility of quantum communications in today's technological landscape is essential. Here, several questions are also raised: are quantum communications feasible in a QFL framework today? What levels of noise do these systems encounter? How might they be vulnerable to side-channel attacks?

Although simulators provide valuable theoretical insights and assist in algorithm development, real-world testing on quantum hardware is indispensable in the current era. This transition from simulation-centric studies to practical experimentation on quantum processors is pivotal for advancing the field and ensuring the viability of QFL in real-world applications.

Adverse trends As discussed in earlier sections, there is a noticeable trend toward prioritizing the study of QFL models' performance. While this focus is undoubtedly promising, an overwhelming number of studies center on binary classification tasks using familiar datasets from classical machine learning, such as MNIST and CIFAR, often in their simplified forms. We contend that there is a significant opportunity to explore a wider range of challenges, such as unsupervised learning problems or pure optimization tasks.

Additionally, a critical issue in the field is the lack of standardization and benchmarking. It is essential to have a consistent framework for assessing the performance of QFL models to facilitate accurate comparisons with other classical or quantum frameworks. Along these lines, the demand for suitable datasets tailored to this area also represents a pressing and unresolved challenge. Developing datasets derived from quantum sensing or other innovative fields could enable more frequent and meaningful engagement with quantum data.

Moreover, other methods and models could be considered, such as the potential advantages offered by quantum annealers, which may provide new avenues for exploration in the field. Similarly, as new classical models in FL emerge continuously, investigating these innovative frameworks through a quantum lens could yield valuable insights and advancements. Bridging these contemporary classical approaches with quantum principles may open exciting research avenues, enhancing the capabilities and effectiveness of FL systems in ways previously unimagined.

Data heterogeneity Data heterogeneity plays a pivotal role in the successful deployment of QFL models, especially when dealing with non-IID data. This factor is crucial not only for performance assessment but also for effective real-world implementation, as actual datasets frequently exhibit non-IID

characteristics. The presence of non-IID data can pose significant challenges, including biased learning and diminished model accuracy, which ultimately undermine the framework's overall effectiveness. In other words, working with non-IID data affects not only model performance but also the reliability and fairness of learned representations. Although a few recent works have begun to tackle this issue, such as by introducing regularization techniques to mitigate client drift, the literature on non-IID data in QFL remains limited and fragmented. As a result, there is still a lack of systematic approaches or benchmarks to evaluate QFL under heterogeneous data conditions. Consequently, the development of QFL models capable of addressing data heterogeneity emerges as a promising avenue for future research. By concentrating on techniques that enhance model robustness and adaptability to diverse data distributions, researchers can significantly improve the practical applicability of QFL across various domains, thereby making it better suited for real-world scenarios where data variability is commonplace.

Device heterogeneity Device heterogeneity is a critical consideration across all three categories, particularly when implementing a QFL model in real-world scenarios. Whether the focus is on optimizing a specific performance metric, deploying frameworks on actual hardware, or enhancing model security, it is essential to account for the varying resources and capabilities of participating clients.

In this context, developing frameworks that dynamically adapt to the computational power and limitations of each client presents a promising research direction. Such adaptability could enhance the overall effectiveness and robustness of QFL systems, ensuring that all clients can contribute meaningfully to the learning process, regardless of their individual hardware constraints.

Inherent quantum advantages Investigating QFL models that offer inherent advantages by design presents an exciting avenue for advancing the field. Rather than adding classical privacy layers, such as DP or HE, there is substantial potential in exploring how QFL models might naturally safeguard privacy and resilience against attacks through their quantum properties. For example, inherent quantum noise or specific quantum data encodings could be harnessed to prevent unauthorized data extraction or eavesdropping, potentially reducing reliance on additional security mechanisms.

Beyond security, it is also promising to examine the expressivity and trainability of QFL models. Many of the known quantum challenges, such as the barren plateau problem, impact a model's ability to converge, limiting its performance and scalability. Research that delves into how QFL models inherently respond to issues like barren plateaus or studies novel quantum encodings that might alleviate these limitations could enhance model robustness.

By focusing on such intrinsic characteristics, we can better understand the unique strengths of QFL and work towards models that are not only powerful but also inherently secure and efficient.

Importance of open-source frameworks Increasing the availability of open-source frameworks in QFL research could significantly accelerate progress in the field. Currently, only a small fraction—approximately 8%—of published studies have publicly accessible code (these frameworks are listed in Table 2). Notably, these works provide concrete implementation examples of QFL using widely adopted quantum simulators and hybrid platforms such as PennyLane, TensorFlow Quantum, TensorCircuit, and Flowr, which support execution on both classical and quantum backends. While most current implementations are tested on classical simulators, many of the frameworks—such as those using PennyLane, TensorFlow Quantum, or TensorCircuit—are designed to be deployable on real quantum hardware via plugin systems or hardware integration interfaces, enabling compatibility with platforms like IBM Quantum (through Qiskit) and Google Quantum (via Cirq).

If more researchers made their frameworks open-source, it would enable the community to evaluate and benchmark models more effectively, facilitating comparisons across different experimental setups and deployment environments.

Additionally, open-source frameworks allow for collaborative refinement and expansion, making it easier to integrate new techniques, adapt models to emerging use cases, and address challenges unique to QFL, such as handling non-IID data.

Open-source code could also play a pivotal role in verifying claims of quantum advantage, as independent researchers could replicate results on different hardware setups and datasets. This transparency not only enhances the credibility of QFL research but also enables more rigorous testing and validation across real-world applications. Ultimately, fostering an open-source ecosystem for QFL would likely drive faster advancements and increase accessibility, benefiting both researchers and practitioners looking to apply QFL in practical scenarios.

Acronyms

<i>BERT</i>	Bidirectional Encoder Representations from Transformers
<i>BQC</i>	Blind quantum computing
<i>DP</i>	Differential privacy
<i>FL</i>	Federated Learning
<i>FTL</i>	Federated transfer learning

Table 2 Publicly available QFL frameworks. Notably, all are implemented on classical simulators

Reference	Framework	Core focus
Ma et al. (2024)	Scikit-learn	Federated quantum
	PennyLane	Kernel learning
Gurung et al. (2023a)	JAX	QFL with
	Tensorcircuit	Blockchain
Dutta et al. (2024)	PTtorch	QFL with
	Flowr	Homomorphic
	PennyLane	Encryption
Zhao (2023)	JAX	Non-IID QFL
	Tensorcircuit	
Li et al. (2023)	PyTorch	QLSTM for
	PennyLane	Language Modelling
Chehimi and Saad (2022)	TernsorFlow	Quantum dataset
	TensorFlow Quantum	
Brei (2022)	TensorFlow	QFL for binary
	TensorFlow Quantum	Classification
Qu et al. (2024)	PyTorch	Fuzzy QFL
	PennyLane	

<i>HE</i>	Homomorphic encryption
<i>HFL</i>	Horizontal federated learning
<i>IID</i>	Independent and identical distributed
<i>IoT</i>	Internet of Things
<i>LSTM</i>	Long short-term memory
<i>NISQ</i>	Noise Intermediate-Scale Quantum
<i>NOMA</i>	Non-orthogonal multiple access
<i>QAOA</i>	Quantum approximate optimization algorithms
<i>QC</i>	Quantum computing
<i>QCNN</i>	Quantum convolutional neural network
<i>QDDL</i>	Quantum distributed deep learning
<i>QDE</i>	Quantum data encoding
<i>QFL</i>	Quantum federated learning
<i>QFS</i>	Quantum feature selection
<i>QKD</i>	Quantum key distribution
<i>QML</i>	Quantum machine learning
<i>QNN</i>	Quantum neural network
<i>QOT</i>	Quantum oblivious transfer
<i>QSDC</i>	Quantum secure direct communications
<i>QSMPC</i>	Quantum secure multi-party communication
<i>URLLC</i>	Ultra-reliable and low-latency communication
<i>VFL</i>	Vertical federated learning
<i>VQA</i>	Variational quantum algorithms
<i>VQC</i>	Variational quantum circuit
<i>VQE</i>	Variational quantum eigensolver

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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